Detecting departure and destination aerodromes from ADS-B

Version 4.2 — detection inside the pipeline

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Summary

Versions 1–3 of this study ran outside the OPDI pipeline. They reimplemented aerodrome proximity with their own coordinate grid, which made every recommendation awkward to adopt: the findings were expressed in nautical miles while production ranks by H3 cell membership, so acting on them meant re-deriving rather than transcribing.

Version 4 moves detection **into the pipeline**. Three modes now share one candidate-detection stage — H3 as the index, exact great-circle distance as the comparison — and are selected by a `mode` parameter on the production function. The pipeline ran end to end on three days of 2025-06, producing a real flight list scored against Network Manager ground truth.

Three results:

Endpoint abstention buys +12.47 pp of departure coverage for -0.84 pp of accuracy. A real trade, not a free win — and a different conclusion from version 3, for a reason worth stating up front.

The detection radius barely matters; the height threshold is everything. Extending the radius from 40 NM to 110 NM changes departure coverage by +0.20 pp. Raising the height cut-off from on-ground-only to unrestricted changes it by more than fifty points. A distance cap costs almost nothing, which is worth knowing because it is the parameter most likely to be tuned.

Preferring aerodromes with scheduled service is close to free. Ranking candidates on a distance penalised for aerodromes without scheduled service gains **+0.67 pp of departure accuracy** and **+1.24 pp of arrival accuracy**, at no measurable cost in coverage.

> **!** The production baseline improved too
>
> Version 3 concluded that endpoint abstention beat production on *both* coverage and accuracy. That no longer holds, and the reason is not that abstention got worse — it is that **the baseline got better**. Running the altitude-trend algorithm on H3 candidates with exact distances lifts its departure accuracy from 97.45% in version 3 to 98.82% here.
> The comparison in this version is therefore harder and fairer than the one before it. Version 3's numbers are not comparable to these and remain published unchanged.

> **i** What changed since version 4
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> Every headline number is unchanged throughout. The point releases report more of what version 4 already computed, plus one new aggregation.
>
> **4.2**
>
> - A breakdown **by class of ground-truth aerodrome** (Section 8.1) — the one new computation. It surfaces the sharpest finding in the study: on small aerodromes and heliports the method is wrong **every single time it answers**, because the detection reference contains large and medium aerodromes only.
> - Overall correctness is reported in the **radius and penalty sweeps**, not only in the results table.
> - The per-aerodrome cross-check extends to the **100 busiest**, and now covers **arrivals as well as departures**.
>
> **4.1**

- Results given for **both roles on all four measures**: arrival overall correctness and both in-area figures were computed but not printed.
- The sweep plotted **on accuracy as well as coverage**, for both roles. Accuracy peaks near the ground and falls as height is admitted, so the coverage view alone overstates the case for a high threshold (Section 5.1).
- The cross-check names the **mode and parameters** it was produced at, and separates the out-of-area bucket from the movement-count ratio it was distorting.
- Figure captions generated from the data rather than written by hand.

1 What the pipeline does before any method runs

The population a method is judged on is fixed long before the method executes. Each stage below discards something, and what survives is what detection sees.

1. Ingest (`ingestion/osn_statevectors.py`). Hourly partitions are read from the OpenSky archive and immediately filtered to the European bounding box `(-25.87, 26.75, 49.66, 70.26)` and decimated to one sample every 5 s. Both filters are applied *before* anything is written. For the three days here that turned roughly 60 GB of raw feed into **142,630,930 state vectors**.

What is discarded: everything outside Europe, and four samples in five. The 5 s decimation is the reason a track's "first sample" can be several seconds into the take-off roll rather than at the stand.

2. Track construction (`pipeline/tracks.py`). State vectors are grouped into tracks by `SHA2(icao24 || callsign)` and split on a gap of over 30 minutes, or over 15 minutes below 5,000 ft. Each sample is also indexed into H3 cells at resolutions 7 and 12.

What is discarded: nothing, but the grouping decides what counts as one flight. Version 2 measured the consequences — roughly one matched flight in ten is split across more than one track, and a few percent of tracks cover more than one flight.

3. Aerodrome zones (`reference/h3_airport_zones.py`). Each aerodrome is surrounded by concentric rings of H3 res-7 cells: 5 NM steps out to 40 NM, then 10 NM steps to 110 NM. Every cell carries the ring band it belongs to, so a consumer chooses its own radius at read time. **33,751,619 (aerodrome, cell) rows** for the 1,353 large and medium aerodromes in the area.

Why it reaches 110 NM: the same table serves ASMA ring crossings at 40 and 100 NM. Detection uses a fraction of it.

4. Endpoint candidates (`pipeline/flights.py`). For each track, the first and last sample are matched against the zone table on `h3_res_7`, and the exact great-circle distance to each candidate aerodrome is computed. The result is cached as `opdi_endpoint_candidates` — **7,491,655 rows**.

This is the step that makes the sweeps in this paper affordable. Deriving candidates from state vectors is expensive; once cached, changing a threshold is a filter and a re-rank over a few million rows rather than a pass over 142 million. Reducing to the two endpoint samples *before* the join is also what makes the full 110 NM reach cheap — about a million rows to match, not the whole month.

i H3 as an index, distance as the comparison

The H3 join answers "which aerodromes are plausibly near this sample" in one equi-join, without a spatial cross product. It does not answer "how near", because a cell inside a 40 NM ring can still be more than 40 NM from the aerodrome. So every candidate is then measured exactly, and the distance — not the cell — decides.

2 How the metrics work

This study turns on the difference between measures that are easy to conflate, so each is defined before any number appears.

Let G be the ground-truth flights and $A \subseteq G$ those the method answered.

Measure	Definition	The question it answers
Coverage	$\lvert A\rvert/\lvert G\rvert$	How often does it say *something*?
Accuracy	$\lvert\text{correct}\rvert/\lvert A\rvert$	When it speaks, how often is it right?
Overall correctness	$\lvert\text{correct}\rvert/\lvert G\rvert$	Of everything, how much did it get right?

A flight whose trajectory was never seen counts against coverage. That is deliberate: ground truth is the denominator throughout, so "we had no data" is a miss rather than an excluded row.

Accuracy alone is close to meaningless here. It says nothing about how often a method speaks, so a method that answers rarely and cautiously can post a better accuracy than one doing far more useful work. Consider two methods on the same flights:

Table 1: Accuracy favours the cautious method; overall correctness shows the eager one labels more than twice as many flights correctly.

Method	Coverage	Accuracy	Overall correctness
Cautious	40.00%	99.00%	39.60%
Eager	95.00%	88.00%	83.60%

Neither is simply better — they suit different uses. A flight list that downstream users treat as authoritative may well prefer the cautious one. The point is that **no single number decides it**, which is why all three are reported throughout.

2.1 Precision and recall, for out-of-area

Some flights have an origin or destination outside the observed area entirely. "Out of area" is therefore a legitimate *answer*, not a failure — and how well a method identifies those cases is measured with the standard pair:

Measure	Definition	The question it answers
Precision	of flights *labelled* out-of-area, the fraction that were	When it says out-of-area, is it right?
Recall	of flights that *were* out-of-area, the fraction it labelled	Of the cases out there, how many did it catch?

Precision is accuracy computed on one label instead of all of them. Recall has no counterpart in the table above, and it is the one that matters here: **high precision with low recall means a method is right when it fires but rarely fires**.

In the standard confusion-matrix vocabulary, with "out-of-area" as the positive class: precision is $TP/(TP + FP)$ and recall is $TP/(TP + FN)$. Coverage has no confusion-matrix equivalent, because it measures whether the method produced *any* label rather than which one.

2.2 In-area coverage and accuracy

Because roughly one flight in twelve is genuinely out of area, a method could raise its apparent accuracy simply by labelling out-of-area liberally. So the same two measures are also reported **restricted to flights whose truth is a real aerodrome** — detection skill with the free wins removed. When headline and in-area figures diverge, the gap is exactly the contribution of out-of-area labelling.

3 The three modes

All three read the same cached candidates. They differ in what evidence they consult and, more importantly, in when they are willing to stay silent.

3.1 `trend` — the current production algorithm

Evidence: every state vector below **FL40** that falls inside an aerodrome's detection zone, currently 30 NM.

Rule: for each (track, aerodrome) pair, smooth altitude over a 5-sample window, count how often the smoothed value rises versus falls, and require a margin of 4 either way. A positive margin is a take-off, negative a landing.

Abstains when: the margin is within ±4 — classified `ambiguous` and **the flight is dropped entirely**.

Failure mode: a flight seen only above FL40 near its origin, or with a noisy altitude trace, silently loses its aerodrome. The FL40 gate is the binding constraint, not the 30 NM radius.

3.2 `nearest` — the *traffic* library's rule

Evidence: the track's first and last sample. Nothing else — no altitude, no ground state, no trend.

Rule: name the nearest aerodrome by effective distance.

Abstains when: never, except when no candidate exists at all.

Failure mode: it answers just as confidently for a track that begins in the cruise as for one that begins at a stand. Its coverage is the highest of any method and its accuracy the lowest, and both facts have the same cause.

3.3 `endpoint` — nearest, with abstention

Evidence: the same two samples, plus their height above **field elevation** and their `on_ground` flag.

Rule: name the nearest aerodrome, but only if the endpoint is within `radius_nm` **and** either below `height_ft` above that aerodrome's elevation or flagged on the ground. Otherwise, if the endpoint lies near the boundary of the observed area, answer out-of-area. Otherwise answer nothing.

Why height above field elevation: a fixed altitude cut-off means nothing at an aerodrome sitting at 5,000 ft. Elevation comes from OurAirports and is missing for about 3% of large and medium aerodromes; where it is missing the height test is skipped and only the ground flag can satisfy the rule, rather than silently comparing against sea level.

Failure mode: it inherits `nearest`'s naming and therefore its confusions between neighbouring aerodromes; the abstention only decides *whether* to answer, never *what* to answer.

4 Results

Table 2: The three modes on 95,116 ground-truth flights, 2025-06-05 to 07. Overall correctness is coverage x accuracy. Produced at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty.

Movements	Mode	Coverage	Accuracy	Overall	In-area cov	In-area acc
departures	trend	68.72%	98.82%	67.91%	74.84%	98.95%
departures	nearest	91.68%	87.78%	80.48%	96.26%	91.17%
departures	endpoint	81.20%	97.98%	79.55%	87.43%	98.41%
arrivals	trend	68.06%	97.87%	66.61%	74.23%	97.94%
arrivals	nearest	88.56%	79.41%	70.33%	91.15%	84.20%
arrivals	endpoint	69.97%	95.10%	66.54%	75.89%	95.45%

`endpoint` sits between the two by construction, and that is the whole design: it keeps `nearest`'s naming rule and adds `trend`'s willingness to stay silent. It answers +12.47 pp more departures than production for -0.84 pp of accuracy.

Whether that trade is worth taking is a policy question rather than a technical one. A flight list treated as authoritative may prefer production's caution; an analysis needing volume may not.

Arrivals remain weaker than departures on every measure, as in all previous versions. A departure's first sample is at or near a stand, with the aircraft stationary shortly before. An arrival's last sample is more often a loss of reception during descent than a landing, and can be anywhere on the approach.

5 What a distance cap costs

The sweep varies the detection radius and the abstention height independently, 90 combinations in all, over the cached candidates.

Both measures are plotted for both roles, because the two axes do not merely differ in strength — they pull in opposite directions, and a coverage-only view hides that entirely.

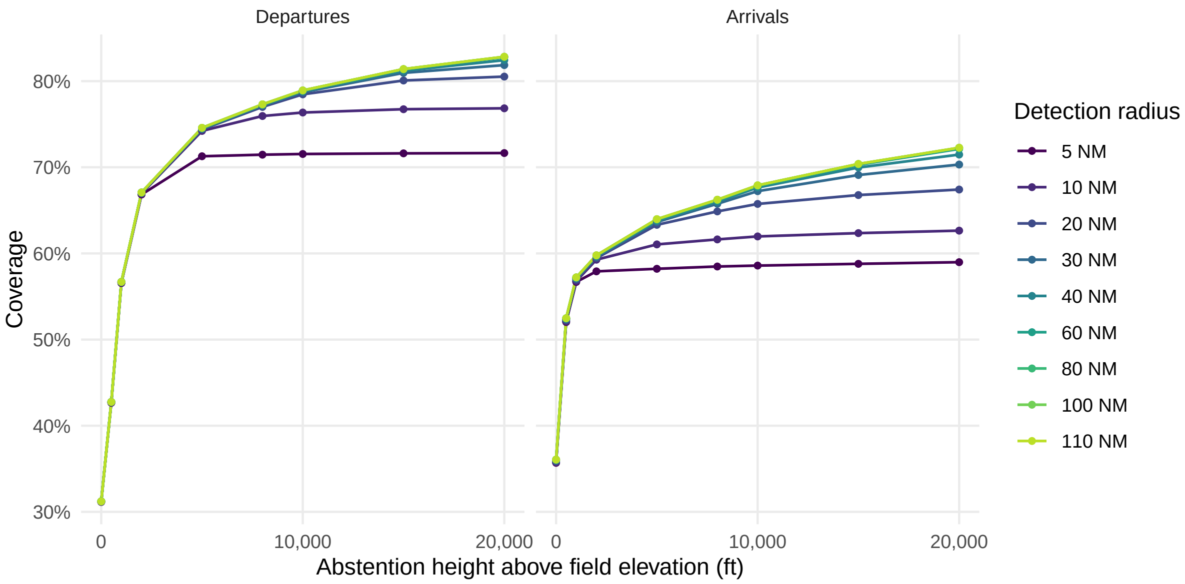

Figure 1: Coverage against abstention height, one line per detection radius. The shape of the curve belongs to the height axis in both panels; the radius only sets how much of it is reachable, and arrivals need more of it than departures. At 15,000 ft a 20 NM radius still leaves +1.32 pp of departure coverage and +3.61 pp of arrival coverage on the table against the full 110 NM reach, while at 40 NM both are within half a point of it.

Table 3: Detection radius at a fixed 15,000 ft abstention height. *ovr* is overall correctness, coverage x accuracy. Coverage saturates by about 30 NM for departures and 40 NM for arrivals; beyond that a wider radius trades accuracy for almost no coverage, and overall correctness turns over.

Max radius	ADEP cov	ADEP acc	ADEP ovr	ADES cov	ADES acc	ADES ovr
5 NM	71.62%	99.07%	70.95%	58.79%	98.64%	57.99%
10 NM	76.74%	98.77%	75.79%	62.36%	97.87%	61.03%
20 NM	80.07%	98.44%	78.82%	66.78%	96.65%	64.54%
30 NM	80.96%	98.16%	79.47%	69.10%	95.63%	66.08%
40 NM	81.20%	97.98%	79.55%	69.97%	95.10%	66.54%
60 NM	81.36%	97.79%	79.56%	70.29%	94.80%	66.63%
80 NM	81.39%	97.75%	79.57%	70.38%	94.69%	66.64%
100 NM	81.40%	97.75%	79.57%	70.38%	94.67%	66.64%

Max radius	ADEP cov	ADEP acc	ADEP ovr	ADES cov	ADES acc	ADES ovr
110 NM	81.40%	97.75%	79.57%	70.38%	94.67%	66.64%

A cap costs almost nothing. Going from 40 NM to the full 110 NM reach changes departure coverage by +0.20 pp and accuracy by -0.23 pp. The useful range is 20–40 NM; below that coverage falls away quickly, above it nothing happens.

Every radius at or above 20 NM traces effectively the same coverage curve: at 5,000 ft they span 74.39% to 74.58%, about a point across a five-fold change. Only the 5 NM and 10 NM lines fall away, and they fall further as the height threshold rises, because a tight radius costs more once more of the sky is admitted.

Height is the parameter that matters. At 40 NM, departure coverage runs from 31.21% with the on-ground-only rule, through 67.05% at 2,000 ft, to 81.20% at 15,000 ft. That is over fifty points from one axis against a fraction of a point from the other.

Arrivals behave the same way with less room: from 36.01% to 69.97% over the same range. They also saturate later in radius — at 15,000 ft a 20 NM cap costs departures +1.32 pp of coverage but arrivals +3.61 pp, because an arrival's last recorded sample is further from the aerodrome than a departure's first.

Two endpoints of the sweep serve as sanity checks, and both behave as they should: at height 0 the rule reduces to "endpoint on the ground" and gives 31.21%, close to what a surface-samples-only method achieved in earlier versions; with height unrestricted and a wide radius it reaches 91.41%, close to `nearest`'s 91.68%, which is what it should converge to.

5.1 What the coverage view hides

Read on accuracy, the same grid tells a different story, and the two together are what actually determines an operating point.

Accuracy peaks near the ground, then declines. At 40 NM, departure accuracy is highest at 2,000 ft (98.98%) and arrival accuracy at 500 ft (98.57%). By 15,000 ft they have given up -1.00 pp and -3.47 pp respectively. Every foot of extra height admits endpoints that are further into the climb or descent, and those are exactly the endpoints most likely to sit nearer a neighbouring aerodrome than the real one.

Arrivals pay roughly three times as much for the same height. That ratio is the single clearest statement of why this study has never recommended changing the arrivals algorithm: the same parameter that is nearly free for departures is expensive for arrivals, so one operating point cannot serve both well.

The radius matters for accuracy even where it does not for coverage. Above 40 NM a wider radius buys essentially no coverage, but it keeps costing accuracy: at 15,000 ft, going from 5 NM to 110 NM costs -1.32 pp of departure accuracy and -3.97 pp of arrival accuracy. This is the real argument for a cap. Version 4 justified one on the grounds that it costs almost no coverage; the accuracy view shows it is mildly *positive* rather than merely harmless.

Overall correctness is what reconciles them. Coverage rises steeply with height and accuracy falls gently, so their product keeps rising for departures across the whole range — which is why 79.55% at 15,000 ft beats 66.36% at 2,000 ft despite the lower accuracy. Arrivals are closer to turning over — 66.54% against 58.70% — which is the same asymmetry again. A user who cares about a trustworthy label rather than a complete one should read the accuracy panel and pick a much lower height than the one recommended here.

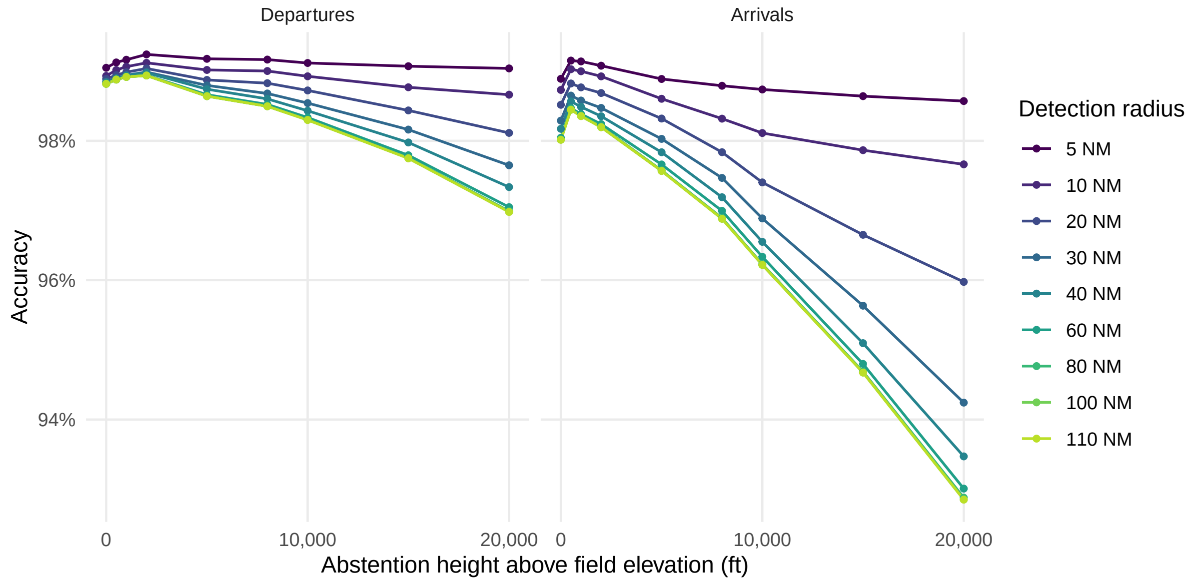

Figure 2: Accuracy against abstention height, same axes. This is the figure the coverage plot hides. Accuracy peaks close to the ground – at 40 NM, departures at 2,000 ft and arrivals at 500 ft – and every foot admitted above that is bought with correctness: -1.00 pp for departures by 15,000 ft, -3.47 pp for arrivals. The radius, which barely touches coverage above 40 NM, spreads accuracy at 15,000 ft by +1.32 pp for departures and +3.97 pp for arrivals.

6 Preferring aerodromes with scheduled service

Version 3 found that the most frequent departure errors formed one pattern: a scheduled civil airport losing to an adjacent military airfield — Izmir to Çiğli Airbase, Naples to Grazzanise, Cagliari to Decimomannu, Catania to Sigonella, Gdańsk to Cewice, all between 7 and 25 NM apart.

The mechanism is straightforward. With 5 s sampling a departing aircraft's first recorded fix is already into the climb and displaced along the departure track. Where a military field lies between the airport and that track, it is genuinely the nearest aerodrome — the rule is not malfunctioning, it is answering the question it was asked.

Aviation knowledge resolves it without new data: a commercial flight almost certainly used the aerodrome with scheduled service. Candidates are therefore ranked on **effective distance** — true distance plus a fixed penalty for aerodromes without scheduled service — so a military field must be *clearly* nearest rather than merely nearest.

Table 4: Scheduled-service penalty at 40 NM and 15,000 ft. *ovr* is overall correctness. A penalty of 0 reproduces the unbiased ranking exactly and is the control.

Penalty	ADEP cov	ADEP acc	ADEP ovr	ADES cov	ADES acc	ADES ovr
0 NM	81.20%	97.31%	79.02%	70.00%	93.85%	65.70%
5 NM	81.20%	97.72%	79.35%	69.99%	94.31%	66.01%
10 NM	81.20%	97.98%	79.55%	69.97%	95.10%	66.54%
15 NM	81.19%	97.90%	79.49%	69.92%	94.88%	66.35%
20 NM	81.20%	97.68%	79.32%	69.91%	94.80%	66.27%
30 NM	81.19%	97.59%	79.23%	69.82%	94.84%	66.22%

The gain is **+0.67 pp of departure accuracy** and **+1.24 pp of arrival accuracy**, for a coverage change of -0.01 pp. Overall correctness moves with it — +0.54 pp and +0.84 pp — which is what makes this the one parameter in the study with no trade attached.

Two features make this trustworthy rather than a lucky constant. The control at penalty 0 reproduces the unbiased ranking, so the effect is attributable. And the curve **peaks and turns** — accuracy is best around 10 NM and declines above it. A penalty that merely favoured larger aerodromes would improve monotonically; one that corrects a specific bias should overshoot, and this does.

> ⚠ It is not free — the cost is a mirror error
>
> Version 3 measured what this preference breaks: genuine movements at non-scheduled aerodromes get pulled to the nearby hub — Biggin Hill to London City, Northolt to Heathrow. In that analysis the mirror error became the **largest single class of remaining departure error**, at roughly 59% of them.
> The net is favourable, and the accuracy figures above include the cost. But a flight list built this way will systematically under-report movements at non-scheduled aerodromes, and anyone analysing general aviation or military traffic should know that before using it.

7 Out-of-area

A flight from Dubai to Amsterdam has a departure aerodrome that no European ADS-B feed can name: the aircraft entered the observed area already airborne. Scoring that as a detection failure understates every method. Treating "out of area" as an answer is an accounting correction, not a detection improvement.

7.1 How it is derived

On the ground-truth side, an aerodrome is relabelled `OOA` when its coordinates fall outside the ingestion bounding box, **or** when its ICAO code does not resolve in OurAirports. The unresolvable codes are overwhelmingly non-European, and a code that cannot be located is one no trajectory endpoint could have been matched to. This is what puts prediction and truth in the same alphabet.

On the prediction side, an endpoint within 30 NM of the bounding box edge is labelled out-of-area. The longitude margin is scaled by $1/\cos(\text{lat})$: a degree of longitude shrinks toward the pole, so an unscaled margin would be roughly twice as strict in northern Norway as in the Canaries.

Precedence is aerodrome first, border second. The opposite order looks equivalent and is not — version 3 found Ponta Delgada, which sits about 8 NM inside the western edge, having its departures labelled out-of-area with the aircraft still on the runway. Being near the boundary means "out of area" only when there is nothing better to say.

The border test carries no height condition. A flight leaving the observed area is at cruise level by definition, so any height cut-off would reject exactly the endpoints the branch exists to catch.

7.2 What it does to the numbers

Table 5: Out-of-area for departures. Only `endpoint` emits the label; the others never do, so their recall is zero by construction.

Mode	truly OOA	labelled OOA	OOA recall	OOA precision	headline acc	in-area acc
trend	8.30%	0.00%	0.00%	0.00%	98.82%	98.95%
nearest	8.30%	0.00%	0.00%	0.00%	87.78%	91.17%
endpoint	8.30%	0.74%	7.94%	89.44%	97.98%	98.41%

`truly OOA` is a property of the sample, not of the method, so it must be identical across all three rows — that it is confirms the decomposition is being computed correctly rather than being a per-method artefact.

The gap between headline and in-area accuracy shows how much out-of-area labelling contributes. It is small here, which is the honest reading: the mechanism is right but the geometric test catches only part of what is there. Version 3 measured this directly at **92–94% precision against 10–15% recall** — when the rule fires it is almost always correct, but it rarely fires. An inbound flight's first recorded sample appears wherever a receiver happens to catch it, which is usually well inside the boundary rather than at it.

The concept is right and the instrument is wrong. A phase-based test would fit the physics better: a track whose first sample is already in the cruise, neither near an aerodrome nor descending toward one, has an origin outside the observed area regardless of where it sits. That is the clearest single improvement available and is not implemented here.

8 Cross-check against the reference

Flight-level accuracy can hide errors that cancel: a departure misattributed from A to B leaves both counts wrong and the total right. Movement counts per aerodrome are what an operational user reads, so they are checked separately.

Table 6: Movement counts against Network Manager, from the `endpoint` flight list at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty. A count ratio below 1 is an undercount. The second ratio drops the out-of-area bucket, whose own recall is poor (Section 7) and which otherwise carries the shortfall of every unlabelled flight.

Movements	Aerodromes	Count ratio	Excl. OOA	Median recall	Median recall, 50+ movements
departures	824	0.810	0.875	80.00%	97.35%
arrivals	805	0.695	0.756	60.00%	84.08%

8.1 By class of aerodrome

Every headline figure averages over populations that behave nothing like each other. The detection reference holds **large and medium aerodromes only**, so flights whose true aerodrome is a small field or a heliport cannot be named correctly at any radius or height. Grouping ground truth by its OurAirports class separates that from detection skill.

Table 7: Coverage and accuracy by the class of the *ground-truth* aerodrome, from the `endpoint` flight list at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty. Classes are OurAirports types. The detection reference contains large and medium aerodromes only.

Movements	Class	Aerodromes	Flights	Coverage	Accuracy	Overall
arrivals	large airport	331	80,683	76.11%	96.27%	73.27%
arrivals	out of area	1	7,964	5.12%	37.99%	1.95%
arrivals	medium airport	346	5,983	74.88%	89.55%	67.06%
arrivals	small airport	70	352	57.67%	0.00%	0.00%
arrivals	heliport	58	134	36.57%	0.00%	0.00%
departures	large airport	331	80,661	88.37%	99.14%	87.60%
departures	out of area	1	7,892	12.34%	64.37%	7.94%
departures	medium airport	352	6,069	78.22%	92.21%	72.12%
departures	small airport	76	361	49.58%	0.00%	0.00%
departures	heliport	65	133	39.10%	0.00%	0.00%

Small aerodromes and heliports are answered confidently and wrongly, every time. Accuracy on both classes is exactly zero, on 49.58% coverage for small airports and 39.10% for heliports. That is not a near-miss rate: the reference cannot name these aerodromes, so when the endpoint happens to fall within 40 NM of some large or medium field below 15,000 ft, the rule names that one instead. Across both roles, 483 of 980 such flights get an answer and none of them are right.

The absolute cost is small — under half a point of the sample — but the shape of the error matters more than its size. **These are not abstentions; they are false statements**, and nothing in the output distinguishes them from the 99.14% of correct large-airport departures. Two fixes are available and neither is implemented here: extend the reference to small aerodromes so they can be named, or make abstention aware of what the reference contains, so an endpoint nearest to an aerodrome the reference omits produces silence rather than its neighbour.

Between the classes the reference does hold, the difference is real but ordinary. Departures at large airports run 88.37% coverage at 99.14% accuracy, against 78.22% and 92.21% at medium ones. Since large airports are 84.80% of the sample, the headline figures are very close to the large-airport figures and should not be read as applying uniformly.

Table 8: The 100 busiest aerodromes by Network Manager departures, against the `endpoint` flight list at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty. Out-of-area is excluded – it is not an aerodrome and is reported separately in Section 7. *Count ratio* is OPDI movements divided by Network Manager movements, so it can sit near 1 even when the individual flights are wrong; *recall* is the share of Network Manager departures OPDI put at this aerodrome, and is the stricter measure.

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LTFM	2,282	2,211	0.97	96.49%
EHAM	2,148	2,165	1.01	99.77%
EDDF	2,094	2,087	1.00	99.43%

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LFPG	2,084	2,071	0.99	98.85%
EGLL	2,001	2,024	1.01	99.35%
LEMD	1,773	1,761	0.99	99.21%
LEBL	1,591	1,582	0.99	99.25%
EDDM	1,573	1,567	1.00	99.43%
LIRF	1,449	1,445	1.00	99.65%
LTAI	1,435	1,387	0.97	96.17%
LEPA	1,417	1,401	0.99	98.80%
LGAV	1,306	1,280	0.98	95.94%
EGKK	1,223	1,218	1.00	99.43%
EIDW	1,185	1,166	0.98	98.14%
LSZH	1,178	1,168	0.99	98.81%
LOWW	1,156	1,142	0.99	98.10%
EKCH	1,140	1,133	0.99	99.30%
LTFJ	1,137	1,115	0.98	96.83%
LPPT	1,010	1,001	0.99	99.11%
LFPO	992	990	1.00	98.99%
LIMC	973	964	0.99	98.56%
ENGM	932	927	0.99	98.93%
EGCC	926	902	0.97	97.41%
EGSS	885	883	1.00	99.66%
EPWA	875	874	1.00	99.43%
EBBR	872	886	1.02	99.66%
EDDB	870	864	0.99	97.70%
LEMG	850	849	1.00	99.65%
ESSA	820	791	0.96	95.98%
EDDL	758	750	0.99	98.15%
LFMN	707	695	0.98	98.02%
LSGG	680	688	1.01	99.41%
LKPR	667	659	0.99	98.20%
EFHK	659	660	1.00	99.09%
LEAL	599	598	1.00	99.83%
LHBP	578	554	0.96	95.67%
EGPH	577	569	0.99	98.61%
EGGW	550	551	1.00	99.82%
EDDH	538	537	1.00	98.14%
GCLP	520	0	0.00	0.00%
EDDK	515	502	0.97	97.28%
LLBG	513	509	0.99	97.66%
LPPR	513	508	0.99	98.83%
LROP	504	493	0.98	97.82%
LEIB	495	490	0.99	98.99%
EGBB	474	468	0.99	98.73%
LIML	458	452	0.99	98.47%
LIME	453	450	0.99	99.34%
LIRN	449	337	0.75	74.61%
LFML	444	440	0.99	98.87%
EDDS	440	418	0.95	94.32%
LGIR	424	1	0.00	0.24%
LIPZ	412	409	0.99	99.03%

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LFLL	411	417	1.01	97.81%
LPFR	405	399	0.99	98.52%
LYBE	394	391	0.99	98.73%
LICC	390	382	0.98	97.69%
LTAC	381	0	0.00	0.00%
LIPE	380	375	0.99	98.68%
LCLK	368	352	0.96	95.38%
EPKK	360	348	0.97	96.39%
GMMN	360	387	1.07	87.78%
EGGD	359	356	0.99	98.89%
LEVC	355	341	0.96	96.06%
LFSB	354	354	1.00	98.59%
ENBR	349	366	1.05	98.28%
LTBJ	346	275	0.79	79.48%
EGPF	344	328	0.95	93.60%
LGRP	340	334	0.98	95.88%
LMML	326	325	1.00	99.39%
GCTS	321	11	0.03	0.00%
GCXO	319	367	1.15	96.24%
LATI	316	0	0.00	0.00%
LICJ	310	288	0.93	92.58%
EBCI	304	304	1.00	100.00%
BIKF	297	331	1.11	98.99%
ELLX	294	292	0.99	99.32%
LFBO	293	291	0.99	98.63%
LGTS	279	271	0.97	96.77%
GMMX	278	88	0.32	30.94%
LEZL	275	273	0.99	99.27%
EVRA	273	271	0.99	98.90%
LTBS	273	219	0.80	80.22%
LBSF	272	169	0.62	62.13%
GCRR	269	0	0.00	0.00%
EGNX	267	265	0.99	98.50%
EPKT	259	249	0.96	96.14%
HEGN	257	0	0.00	0.00%
UGTB	257	199	0.77	61.87%
LFRS	256	254	0.99	98.83%
EDDP	256	257	1.00	99.61%
EPGD	248	39	0.16	15.73%
LGKR	244	212	0.87	84.84%
HECA	244	42	0.17	17.21%
EGAA	239	240	1.00	100.00%
ENZV	236	174	0.74	67.80%
LEBB	234	174	0.74	73.50%
LFPB	230	230	1.00	98.70%
LIEO	228	195	0.86	85.53%
LIBD	228	0	0.00	0.00%

Table 9: The 100 busiest aerodromes by Network Manager arrivals, against the `endpoint` flight list at 40 NM detection radius, 15,000 ft abstention height, 10 NM scheduled-service penalty. Out-of-area is excluded – it is not an aerodrome and is reported separately in Section 7. *Count ratio* is OPDI movements divided by Network Manager movements, so it can sit near 1 even when the individual flights are wrong; *recall* is the share of Network Manager arrivals OPDI put at this aerodrome, and is the stricter measure.

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LTFM	2,258	1,647	0.73	72.85%
EHAM	2,136	1,766	0.83	82.49%
EDDF	2,091	1,609	0.77	76.66%
LFPG	2,066	1,546	0.75	74.54%
EGLL	1,986	1,183	0.60	57.85%
LEMD	1,786	1,357	0.76	74.92%
LEBL	1,597	1,411	0.88	88.17%
EDDM	1,571	1,313	0.84	83.45%
LTAI	1,437	994	0.69	68.89%
LEPA	1,436	1,386	0.97	96.24%
LIRF	1,433	1,189	0.83	82.76%
LGAV	1,317	1,128	0.86	83.45%
EGKK	1,220	994	0.81	81.39%
EIDW	1,188	965	0.81	81.14%
LSZH	1,160	978	0.84	83.97%
EKCH	1,152	1,007	0.87	87.41%
LOWW	1,145	896	0.78	78.17%
LTFJ	1,138	925	0.81	80.67%
LPPT	1,001	764	0.76	76.32%
LFPO	991	888	0.90	89.51%
LIMC	967	757	0.78	77.56%
ENGM	940	877	0.93	92.66%
EGCC	924	619	0.67	66.99%
EGSS	888	773	0.87	86.49%
EPWA	880	705	0.80	79.77%
EBBR	871	733	0.84	83.47%
LEMG	861	789	0.92	91.64%
ESSA	841	709	0.84	83.83%
EDDB	840	763	0.91	89.88%
EDDL	751	666	0.89	88.15%
LFMN	710	657	0.93	91.97%
LSGG	670	625	0.93	91.64%
EFHK	666	577	0.87	85.59%
LKPR	657	571	0.87	86.45%
LEAL	599	572	0.95	95.49%
EGPH	584	506	0.87	85.96%
LHBP	577	238	0.41	41.07%
EGGW	552	479	0.87	86.41%
EDDH	532	488	0.92	90.04%
GCLP	521	0	0.00	0.00%
LLBG	516	409	0.79	74.03%
LPPR	512	435	0.85	84.77%
EDDK	509	443	0.87	85.66%

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
LEIB	509	493	0.97	96.66%
LROP	509	463	0.91	90.77%
EGBB	466	388	0.83	83.26%
LIML	465	445	0.96	94.62%
LIME	460	419	0.91	91.09%
LIRN	447	335	0.75	73.60%
LFML	445	391	0.88	87.64%
EDDS	433	377	0.87	86.84%
LGIR	430	0	0.00	0.00%
LFLL	412	386	0.94	90.53%
LIPZ	409	370	0.90	89.98%
LPFR	407	392	0.96	96.31%
LYBE	395	342	0.87	86.33%
LICC	391	356	0.91	91.05%
LIPE	376	359	0.95	95.21%
LTAC	374	1	0.00	0.00%
LCLK	367	338	0.92	91.55%
EGGD	360	303	0.84	84.17%
ENBR	357	356	1.00	93.00%
GMMN	357	336	0.94	71.99%
LEVC	357	333	0.93	93.00%
EPKK	356	250	0.70	70.22%
LFSB	352	322	0.91	91.48%
LGRP	346	312	0.90	87.57%
EGPF	346	299	0.86	85.55%
LTBJ	341	275	0.81	80.65%
LMML	326	310	0.95	94.79%
GCTS	323	97	0.30	21.98%
LATI	323	0	0.00	0.00%
GCXO	320	316	0.99	23.75%
LICJ	314	291	0.93	92.04%
EBCI	303	285	0.94	94.06%
BIKF	296	165	0.56	29.39%
ELLX	294	226	0.77	76.53%
LFBO	290	278	0.96	95.17%
LGTS	281	264	0.94	93.95%
LTBS	280	228	0.81	81.07%
GMMX	279	57	0.20	19.35%
EVRA	278	221	0.79	78.78%
LEZL	276	255	0.92	92.39%
GCRR	269	0	0.00	0.00%
LBSF	267	89	0.33	32.58%
EDDP	261	224	0.86	84.67%
HEGN	258	0	0.00	0.00%
LFRS	257	237	0.92	92.22%
EPKT	257	211	0.82	82.10%
UGTB	257	101	0.39	21.79%
EGNX	255	206	0.81	79.61%
LIEO	252	212	0.84	83.33%
LGKR	247	222	0.90	68.02%

Aerodrome	NM movements	OPDI assigned	Count ratio	Recall
EGAA	246	223	0.91	90.65%
EPGD	244	10	0.04	1.64%
ENZV	236	130	0.55	46.61%
LEBB	235	53	0.23	16.17%
LTFE	232	187	0.81	80.17%
HECA	231	82	0.35	34.20%
LIBD	229	1	0.00	0.00%

The error is an **undercount** rather than an overcount, which is the correctable direction: a systematic shortfall can be adjusted for, an inflated count cannot. Roughly six points of it are the out-of-area bucket rather than per-aerodrome detection failure — flights the method never labelled land nowhere, so they show up as missing movements at whichever aerodrome the truth names.

Recall also scales sharply with aerodrome size. Across the hundred busiest — 74.96% of all departures in the sample — the median is 98.24% and the quartiles are 95.60% and 99.04%, against a median of 80.00% over all 824 aerodromes with departures. The flight list is therefore substantially more complete at large aerodromes than small ones, which matters before comparing across aerodrome classes — and Section 8.1 says why.

8.2 Busy aerodromes that fail outright

Extending the listing from thirty rows to a hundred exposes something the shorter view could not: a handful of genuinely busy aerodromes where the method does not merely lose a few movements but fails almost completely.

29 of the two hundred rows above sit below 50% recall — 11 departure aerodromes and 18 arrival ones. They split cleanly into two kinds of failure:

- **Never seen** (19 rows). Under a quarter of their flights get an answer at all: Gran Canaria departures return 24 answers against 520 movements, and Hurghada none at all. These cluster where ADS-B receiver coverage is thin — the Canaries, North Africa, the Caucasus, Anatolia. No detection rule can recover a flight with no low-altitude trajectory.
- **Seen and misnamed** (10 rows). The trajectory is there and the answer is wrong. Bilbao arrivals are answered 90.64% of the time and only 38 of 235 land on Bilbao.

Tenerife North is the cleanest warning about reading count ratios. It is assigned 316 arrivals against 320 true ones — a count ratio of 0.99, which looks like a well-measured aerodrome — while its recall is 23.75%. Almost every movement it is credited with belongs somewhere else, and the totals cancel the error exactly as Section 8 warned they could.

Every aerodrome in both groups is a `large_airport` with scheduled service and is present in the detection reference, so neither failure is the reference gap of Section 8.1. The first group is a data limitation and is not fixable here. The second is a detection failure at aerodromes large enough to matter, it is concentrated rather than scattered, and **this study has not diagnosed it** — it is the most promising lead the cross-check produces.

9 Recommendation

Adopt `endpoint` for departures if coverage matters more than the last point of accuracy. It adds +12.47 pp of coverage for -0.84 pp, and it is a smaller algorithm than the one it replaces — no smoothing window, no margin parameter, no per-aerodrome vote. If accuracy is paramount, keep `trend`; it is genuinely the more accurate rule now.

Use 20–40 NM and a scheduled-service penalty of 10 NM. Both are on plateaus, so neither is delicate.

Do not change the arrivals algorithm on this evidence. Arrivals are weaker on every measure and every version of this study has found the same.

Publish the provenance column. `adep_source` / `ades_source` distinguish "named an aerodrome", "outside the observed area" and "could not determine". Today all three are a single null in the published flight list, and they mean entirely different things to a consumer. This is the highest-value change per line of code in the whole study.

Make abstention aware of what the reference contains. Flights to and from small aerodromes and heliports are answered about half the time and are wrong every time (Section 8.1). They are a small share of movements but they are false statements rather than gaps, and they are indistinguishable in the output from correct answers. Either extend the reference to small aerodromes or abstain when the nearest candidate is far enough away to suggest the real aerodrome is one the reference does not hold.

10 Limitations

- **Three days**, one sample. Versions 2 and 3 used two samples a year apart; this version re-runs the whole pipeline, which is expensive, so it uses one. The mode ranking is robust across earlier versions, but the exact figures here have not been validated on a second period.
- **Out-of-area recall is low.** The geometric border test is the wrong instrument; the phase-based test in Section 7 is the successor.
- **The detection reference holds large and medium aerodromes only**, so accuracy on any other class is zero by construction rather than by measurement. The per-class table quantifies this; it does not fix it.
- **The scheduled-service preference has a cost** that these accuracy figures include but do not itemise: movements at non-scheduled aerodromes are pulled to nearby hubs.
- **The sweep re-implements the endpoint rule** as a DataFrame operation instead of calling the pipeline's `classify_endpoints`, because a sweep cell needs only `(track_id, adep, ades)` while the pipeline method builds a whole flight list. The headline `endpoint` row *does* come from the pipeline path, so a divergence between the two would surface as the sweep disagreeing with its own mode — but this is a duplication and worth removing.
- **`track_id` is taken as given**, with the fragmentation and merging rates version 2 measured.

11 Reproducing

Reference zones, state vectors and tracks, in the `opdi` repository:

```
RANGE="--env opensky --start 2025-06-05 --end 2025-06-08"
opdi run $RANGE --step 00a   # zones: 5 NM bands to 110 NM
opdi run $RANGE --step 01    # state vectors, bbox and 5 s applied at ingest
opdi run $RANGE --step 02    # tracks, with H3 res 7 and 12
```

Candidates once, then a flight list per mode:

```python
from datetime import date
from opdi.pipeline.flights import FlightListProcessor

fp = FlightListProcessor(spark, config)
fp.build_endpoint_candidates(date(2025, 6, 1))
for mode in ("trend", "nearest", "endpoint"):
    fp.process_dai(date(2025, 6, 1), mode=mode, skip_if_processed=False,
                   abstention_radius_nm=40, abstention_height_ft=15000,
                   sched_penalty_nm=10,
                   table_name=f"research/flight_list_{mode}",
                   write_mode="overwrite")
```

Scoring and both sweeps. This runs locally: the inputs are small once the candidates are cached, and the cluster namespace is shared.

```
python benchmarks/benchmark_modes.py --months 202506 \
    --days 2025-06-05 2025-06-06 2025-06-07 --local --results-dir <dir>
```

Rendering this page needs no credentials, no database and no cluster.